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A Recurrent Neural Network-based Multi-Parameter Sub-Band, Sub-Second Analysis For Voice Spoofing and Cloning Detection

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***Abstract*—Speech synthesis, voice conversion, and replays are all forms of voice faking that might compromise Automatic Speaker Verification (ASV) systems. The ability to manipulate audio is a significant tool that criminals may use to take control of smart homes, bank accounts, and more. To detect Logical Access and Physical Access spoofing attacks using Multi-Pattern Features, this study employs a modified RNN architecture and Softmax layer. A robust voice spoofing detection system that can detect many types of spoofing attacks, intending to prevent ASV system fraud, has been provided. Local Binary Pattern Histogram (LBPH) is a novel feature descriptor that is provided for audio representation. By evaluating audio in both directions, LBPH can detect artificial speech artifacts, distortions in replay microphones, and dynamic speech characteristics in the real signal. Replay and logical-access attacks, such as speech synthesis and voice conversion, are detected by the Recurrent Neural Network (RNN) by using the recommended LBPH features. RNN is used since it is better at digesting and learning the representation of sequential input. It is discovered that physical-access attacks had an equal error rate of 0.69% and logical-access attacks with an accuracy of 90%. In addition to classifying speech cloning techniques, the proposed system can detect undiscovered voice spoofing attacks. When tested on the ASVspoof 2019 corpus, the suggested approach outperforms the state-of-the-art voice spoofing detection algorithms in detecting physical and logical access attacks.**

Keywords—Speech Synthesis, Voice Conversion, Automatic Speaker Verification, Smart Homes, Bank Accounts, Voice Spoofing Detection, Local Binary Pattern Histogram, Logical Access, Physical Access, Equal Error Rate, Recurrent Neural Network

I. Introduction

Automatic Speaker Verification systems (ASVs) have grown very popular since the COVID-19 pandemic. Physical contact is necessary for many identity authentication techniques, including fingerprint scanning and passwords. Due to the worldwide scope of these health issues, it is essential to use methods that do not require human intervention. Many individuals constantly use ASV, whether it's for banking, operating their watches, or making phone calls. The microphone sends the audio signal to a speech-to-text (STT) system, which utilizes it to determine the answer. Based on the result, it then determines whether or not to accept or reject the voice. Determining the validity of an input's audio is the main objective of an ASV system. ASV-based technologies make lives easier, but they also present several difficulties. An unauthorized user trying to log in by mimicking the voice of a genuine user is one possible danger to ASV systems.

On an iPhone, either Google Home or Siri can do a lot of things, such as check the weather, make calls, access the web, set reminders, and unlock the phone [1]. Since the COVID-19 epidemic has made fingerprint scanning and password-based verification techniques illegal, ASV-based authentication is becoming more popular. Voice biometrics is becoming more popular for user identification. Vocal spoofing attacks, including replays, synthesis, voice conversion (VC), and user authentication, may affect ASV systems. Houses and bank accounts are among the many items that may be taken in such attacks. Hackers have recently used voice spoofing to conduct fraud by impersonating ASV systems. Theft from a concealed account was the result of a US lawsuit where the CEO's speech was used as a bargaining chip [2]. Scientists are hard at work on robust voice anti-spoofing technologies to shore up ASV systems and prevent speech-spoofing attacks.

Numerous apps and gadgets may now use biometric identification thanks to modern technology. The emphasis is the need for measures to avoid identity fraud due to the widespread use of synthetic media. Because they rely only on sound detection, ASV systems in telephony are vulnerable to fraudulent authentication efforts [3]. Numerous neural network models have been created to detect audio that is meant for identity theft. Primitive neural networks were first used to accomplish this goal. No matter how far technology came, increasingly complex designs were applied to make this attempt more efficient.

Voice-spoofing techniques might be useful in protecting against logic-access (LA) and physical-access (PA) threats. The use of inverted mel-frequency cepstral coefficients (LPCC), IMFCC, and LPCC to evaluate high-frequency band audio representation [4]. The Gaussian mixture model (GMM) was trained to distinguish between replay and real data using these three spectral features. To study the extended Constant-Q cepstral coefficients (eCQCC), [5] converted an octave power spectrum to a linear one using the constant-Q transform and fixed resampling. The voice spoofing detection method in this research works for PA and LA attacks. LBPH is introduced to help distinguish authentic from counterfeit audio samples. LBPH features were used to train the RNN for

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classification. RNNs are useful for time-series data processing, hence they are utilized for categorization. Additionally, the recommended approach may accurately classify the cloning algorithms utilized to manufacture valid samples. The ASVspoof 2019 corpus was used to test the approach on two types of data: genuine and replay samples, speech synthesis, voice converter, and legit LA samples. The following are the main contributions.

- New acoustic LBPH features that can accurately identify authentic and fake samples by extracting information in both directions from audio signals, is proposed.
- The speech spoofing detection method is reliable and secure against logical and physical access attacks alike.
- The method for detecting spoofing can even pick up on subtle synthetic speech attacks.
- A system that can identify the algorithm used to create realistic sound effects, is developed.
- The spoofing detector is put through its paces on the ASVspoof 2019 corpus, along with other methods, to gauge its performance.
- A straightforward pre-processing phase that subsamples just spoken frames before spoofing detection, is proposed.
- While reducing the number of necessary data by a factor of four in this pre-processing stage, overall detection accuracy (same pooled EER) is maintained.
- This pre-processing step is shown to be compatible with many current spoofing detection methods.
- The detection accuracy for the challenging ASVspoof 2019 A17 voice conversion attack is proved to be improved by integrating two separate detection approaches with the novel pre-processing phase.
- Three pattern images are combined to detect audio spoofing. Modifying DenseNet allowed to extract pattern images from Mel spectrograms.
- After utilizing Softmax to assess the current pattern image at the end of each network, three scores are used to categorize the input audio. This fusion decision may improve the model's performance, according to the experiments.

II. Related Works

This implies that CQCC based on CQT magnitude may omit phase information that varies during playback. CQTMGD, a modified group delay feature, determines the CQT phase. They identified input signals via a multi-branch residual convolutional network. CQTMGD outperforms MGD. Das, Yang & Li (2020) discovered that eCQCC and CQSPIC identify fake input, improving reliability [6].

Experimental data show the proposed technique enhances baseline works. To create subband transform signal encoding. Constant Q equal, octave and mel subband modifications were studied using ASVspoof 2015 and 2019 logical access statistics. Experiments show that subband-based features outperform band transforms. The 2021 Modified magnitude-phase spectrum (MMPS) feature by Yang et al. measures input signal amplitude and phase. CQT-MMPS is a custom MMPS feature with constant Q transform.

CQMCC surpassed most spoofing detection algorithms. It used frequency band calibration, sinc convolution, and squeeze-excitation modules to obtain temporal waveform features. Method improved covert attack detection generalization [7]. To suggest sm-ALTP characteristics to detect audio spoofing fraud. Weak classifiers form a stable classifier. Their method was tested on ASVspoof 2019 and VSDC. CS-LBP, a novel feature descriptor [8, 9], optimizes audio file encoding. For audio counterfeit identification, the long-short-term memory network received these properties. In 2022, Li et al. showed that long-term relations and high-frequency data may identify misleading signals. They suggest the L-VQT for counterfeit signal detection [10].

Improving self-supervised learning for audio spoofing detection can be achieved by addressing the challenges in embedding fusion using layer-conditioned techniques. Traditional models often fail to effectively integrate multi-layered representations, limiting their ability to generalize across varied spoofing attacks. Layer-conditioned embedding fusion allows the model to dynamically weigh and combine features from different network layers, preserving both low-level and high-level information. This method enhances discriminative power and robustness, particularly in detecting subtle artifacts in spoofed audio. Techniques such as attention-based fusion, gating mechanisms, or learnable conditioning vectors can be employed to optimize fusion, resulting in improved performance across diverse spoofing scenarios [11].

DeepDet, which integrates YAMNet with a Bottleneck Attention Module (BAM), aims to enhance TTS synthesis detection by addressing limitations in capturing fine-grained acoustic features. A key challenge is that standard models often overlook subtle artifacts in synthetic speech, reducing detection accuracy. By incorporating BAM into YAMNet, DeepDet enhances the model's focus on relevant spectral-temporal regions, improving its sensitivity to synthesis artifacts. The bottleneck structure ensures efficient parameter usage, while the attention mechanism enables dynamic feature refinement. This method strengthens the model's ability to differentiate between natural and synthetic speech, offering robust detection across various TTS systems and challenging real-world scenarios [12].

Speech deepfake detection faces significant challenges due to the rapid advancement of generative models like TTS and voice conversion systems, which produce highly realistic synthetic speech. Key problems include generalization across unseen attacks, variability in recording conditions, and limited availability of diverse training data. Detection methods broadly fall into feature-based and model-based approaches. Feature-based methods extract spectral, prosodic, or phase-related cues to distinguish fakes, while model-based techniques leverage deep learning architectures such as CNNs, RNNs, and transformers. Recent trends emphasize self-supervised learning, domain adaptation, and ensemble models to improve robustness and scalability, aiming to keep pace with evolving deepfake technologies [13].

III. Proposed Work

The suggested defense against voice spoofing here is discussed. The novel feature extraction approach, LPBH, is

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also detailed in length. A ten-layer Recurrent Neural Network is trained utilizing the proposed LBPH features to differentiate between real and fraudulent audio. To accurately classify authentic and fraudulent audio, it is critical to identify and extract pertinent information from them. The study demonstrates that spectrograms and mel-frequency cepstral coefficients combine to surpass the existing gold standard. The methodology is shown forth in Fig. 1

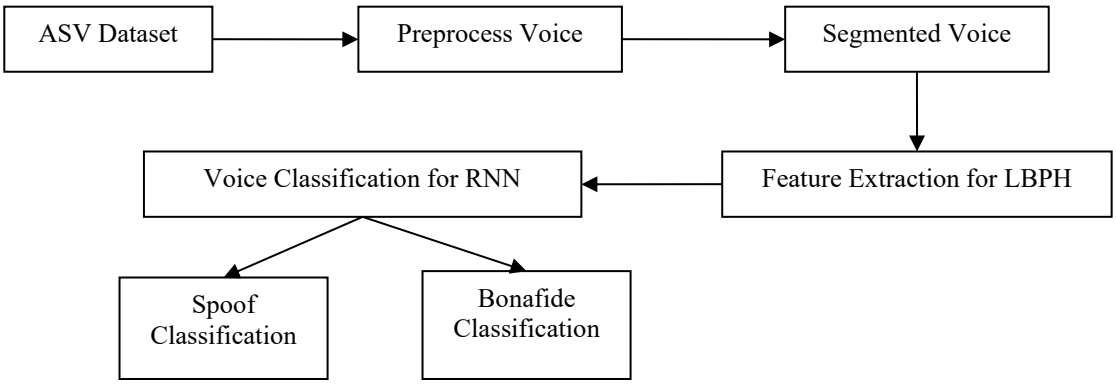

Fig. 1. System Architecture

A. ASV Dataset

2015 ASVspoof events highlighted the need for spoofing detection tools. Because they were easier to implement in the absence of technology, replay attacks were the focal point of the second spoof challenge of 2017. Replay attacks, voice conversion, and speech synthesis were all part of the 2019 third spoof challenge. In the fourth competition of 2021, ASVspoof 2019 was used to detect deepfake and fake voices. Partition LA contains samples of voice conversion and speech synthesis in the ASVspoof 2019 challenge database, whereas Partition PA stores replay. There is training, growth, and evaluation in every part. Model detection is assessed in the evaluation set, whereas development and training are tested in separate subsets. As part of their training and growth, LAs undergo six rounds of techniques. The model is put to the test against dangers it was not trained on in eleven unidentified LA attacks. There is a human-generated voice for every grouping. This speech was inspired by an actual VCTK corpus utterance [14]. A detection model is constructed by dividing the 107 speakers' utterances into 46 male and 61 female categories.

B. Voice Pre-process

Using the top ASVspoof 2019 scores (excluding ensemble systems) and two independent detectors, the suggested pre-processing step is evaluated. By using this approach, more traditional systems are pre-processed instead of relying on deep learning and neural networks. The first two systems employ a GMM-based ML detector with LFCC features [15] and IA/IF statistics [16] from deconstructed IMFs using EMD. Consider the pre-processing phase of the anti-spoofing countermeasure to be its front end. The front end will only save spoken frames after a two-fold subsampling of the voice stream. Consequently, its front end reduces processing time by half by sending less data to the countermeasure. Spoken frames make up half of the speech signal, while silent or unvoiced frames make up the other half.

C. Voice Segmentation

Earlier methods for detecting falsified speech relied on pre-trained algorithms that could distinguish between genuine and false signals by extracting unique characteristics from the whole audio stream. Feature extraction from deconstructed speech signals including more discriminating information was the subject of subsequent research to improve detection accuracy. Two spectral variants in voiced and unvoiced segments from real and spoof speech, including two female speakers (LA79 and LA80), two male speakers (LA92 and LA95), and ASVspoof 2019 LA training sets A01-A04 text-to-speech (TTS) are considered. Then, the real and fake versions of the same line said by each speaker can be distinguished. The next step is to classify each matched phrase's phonemes as either voiced or unvoiced. Lastly, the difference in magnitude spectrum between the real and false versions was shown by both the spoken and unvoiced sections. Based on the observations on the spectrum differences in voiced segments between real and fake speech, a pre-processing phase that receives the speech signal and sends out a signal containing just voiced segments to the countermeasure is proposed. The bulk of the spectral dissimilarity in the 0-4 kHz range is present in the voiced segment, hence double-subsampling the signal is another alternative.

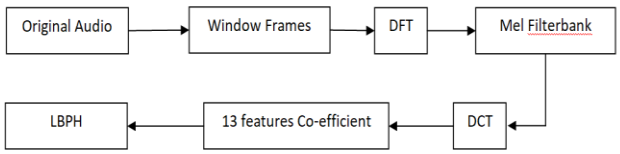

Fig. 2. LBPH Flow Diagram

D. Feature Extraction for LBPH

Samples are gathered over some time, and the LBPH (see Figure 2) determines their frequency content to produce spectrograms. Many signal frames are retrieved and examined at each window. Every frame updates a matrix that represents the energy and spectrum of the signal. The array's first slot is filled with new frames one after the other. As a result, the energy and signal spectrum may be shown as a function of time. The most extensively reported spoof detection technique in specialized literature is the mel frequency cepstral coefficients feature extraction approach. Before LBPH is retrieved, analog signals are digitalized by sampling them at a predetermined rate. Numerous processes are used for digital signals. To get the Discrete Fourier Transform (DFT) of the analog signal, divide it into overlapping frames. After the signal is filtered by the Mel filter bank, the output is log compressed. The Discrete Cosine Transform (DCT) is then

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used to transform this into the cepstral domain, keeping the first 13 coefficients and discarding the higher ones. In this study, LBPH algorithms are used. First, a 13 × 298 matrix is created by extracting 13 coefficients for every window. The matrix is reduced to a vector of 13 points after the average of all windows has been calculated. Matrix and vector are used for audio classification.

E. RNN Voice Classification

With a binary cross-entropy loss function, an RMSprop optimizer, 200 batches, and four epochs, the linear spectrogram adaptation is achieved. In comparison, using the logarithmic spectrogram variation in conjunction with the same loss function and Adam optimizer required 10 iterations and twenty batches. The batch size was also reduced. The model takes the mel frequency vector as input and uses a dropout layer, three time-distributed one-dimensional dense layers with 24, 13, and 10 units each, and finally outputs the 13 coefficients. After that, the process is completed by adding a second dropout layer and three LSTM layers with 10, 15, and 30 additional units, respectively. In the end, a 13-coefficient unit thick layer was added. The Equal Error Rate (ERR) attained was 0.69% lower than the ERR that was obtained.

IV. Experimental Results

This section details the experiment that was run to see how well the suggested system could identify voice spoofing by applying the suggested method to the ASV dataset's PA and LA sets starting in 2019, along with the following metrics: EER, recall, accuracy, and precision. For the sake of implementation, Anaconda Python was used. In addition, a computer system that met particular requirements was used i.e., computer has a Core i57 processor and 12 GB of RAM.

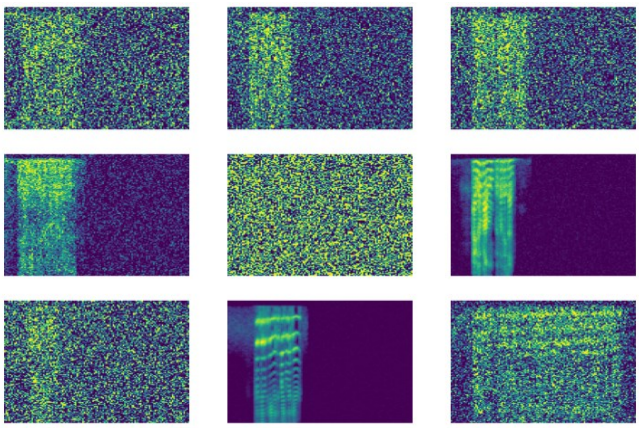

Fig. 3. Preprocess Image

Figure 3 shows an example of a spectrogram, which is a visual representation of the time-frequency components of an audio source. All of the plots show the changes made to the audio data during preprocessing, which could highlight features like segment separation or noise reduction. Spectrograms have several machine learning-related uses, such as audio classification and speech recognition.

There is a time-varying audio stream shown by the waveform up there. In human speech, the 0-4 kHz range is necessary, and the voiced segments are characterized by denser wave peaks that include these frequency components. The ability to reduce background noise and detect and remove unvoiced sections makes it possible to evaluate essential data for applications like speech processing, which is made possible by separating these regions shown in above figure 4.

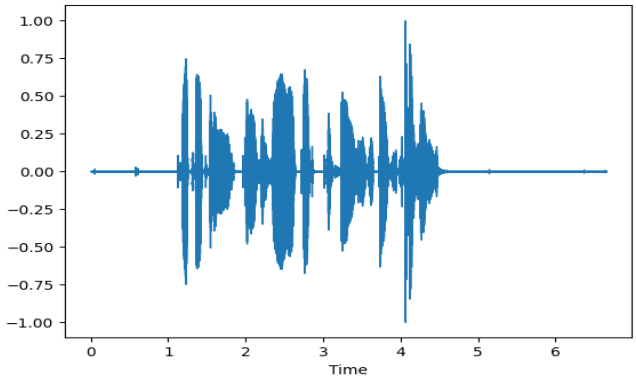

Fig. 4. Voice Segmentation

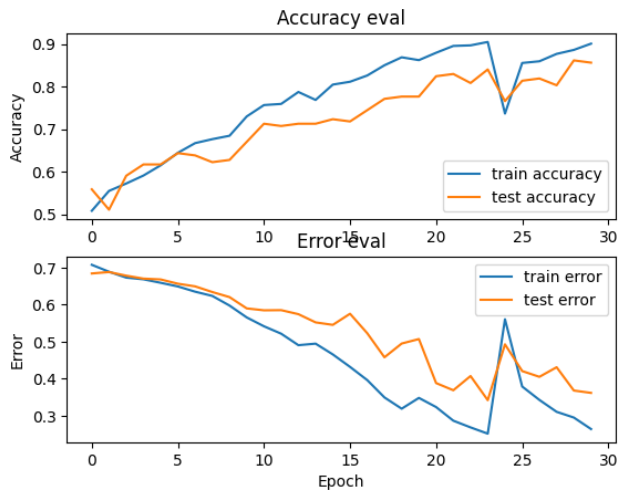

Fig. 5. Accuracy Result

The figure 5 displays the trends of testing and training for speech classification, including spoof and real detection. With a 90% accuracy rate and a 0.69% error rate, the model demonstrates exceptional ability in distinguishing between genuine and false sounds. Efficacious learning and robust feature discrimination are shown by the dataset over time, as the error lowers and the accuracy grows.

V. Conclusion

To counter various forms of LA and PA spoofing attacks, this research has developed a robust speech spoofing detection system that combines the novel LBPH features with LSTM.To effectively include the attributes of microphone distortions in replay signals, anomalies from the cloning process, and authentic speech dynamics, a distinctive feature representation method for LBPH is developed. The proposed approach effectively distinguishes various types of voice spoofing techniques, as shown in trials with a comprehensive and diverse ASVspoof 2019 corpus. The method achieved an Equal Error Rate (EER) of 0.69% for LA attacks and 0.67% for PA attacks. The cloning algorithms used to generate the synthetic voices may be identifiable by the technology. The analysis indicates that the speech spoofing detection technique surpasses the most sophisticated alternatives now available. The ASVspoof corpus evaluation set comprises hitherto unencountered speaker data; please consider this. The remarkable results obtained from the ASV Spoof 2019 corpus

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evaluation set demonstrate the efficacy of the cross-dataset assessment methodology is an accuracy of 98%. In future research, the approach can be evaluated in a cross-dataset context using two distinct voice synthesis datasets.

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